Instrumental Variables

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September 18, 2026

Introduction

Instrumental variables (IV) methods have been at the center of applied microeconometrics for decades

- Pioneered by economists and now widely used in other empirical fields

We start with constant effects, then reinterpret IV with heterogeneous effects

IV history

IV methods were developed to deal with two seemingly unrelated problems

- Philip and Sewall Wright (1920s): estimating supply and demand from equilibrium prices and quantities
  - Grew into the study of simultaneous equations models
- Wald (1940): estimating a regression line when the regressor is measured with error

Today IV is mostly used to eliminate omitted variables bias in single-equation causal models

Outline

Part I covers IV with constant treatment effects

- Simple IV: one instrument, one endogenous variable
- Multiple instruments and 2SLS
- IV as GMM
- Overidentification tests
- Imperfect compliance
- 2SLS mistakes
- Many/weak instruments

Part II covers IV with heterogeneous treatment effects

- The LATE theorem
- Characterizing compliers
- Overidentification tests with heterogeneous effects

Main reference: Mostly Harmless Econometrics

Section 1

Part I: Constant effects

A simple causal model

Consider a constant-effects causal model of schooling and earnings:

$$Y_i(s) = \alpha + \rho s + \varepsilon_i$$

- $Y_i(s)$ is i's potential wage with s years of schooling
- Additive separability implies constant causal effects:

$$Y_i(s) - Y_i(s-1) = \rho \quad \forall\,(i, s)$$

We observe schooling S_i and earnings $Y_i(S_i)$

Observed earnings and unobserved ability

Observed earnings follow the structural equation

$$Y_i = \alpha + \rho S_i + \varepsilon_i$$

- This is not a regression, so nothing guarantees $Cov(S_i, \varepsilon_i) = 0$

Project ε_i onto unobserved ability A_i:

$$\varepsilon_i = \gamma A_i + v_i$$

- $Cov(A_i, v_i) = 0$ by definition, and we assume $Cov(S_i, v_i) = 0$
- Unobserved ability is a potential source of omitted variables bias

Long and short regressions

The long regression controls for ability, and the short regression omits it:

$$\begin{aligned} \text{Long:} \quad & Y_i = \alpha + \rho S_i + \gamma A_i + v_i \\ \text{Short:} \quad & Y_i = \alpha_s + \rho_s S_i + v_{is} \end{aligned}$$

The omitted variables bias formula links the two:

$$\rho_s = \rho + \gamma \frac{Cov(S_i, A_i)}{Var(S_i)}$$

The regression we've got (short) is not the regression we want (long)

IV recovers the long regression coefficient ρ without data on A_i

Instrumental variables

Consider an instrument Z_i for S_i in the causal model

$$Y_i = \alpha + \rho S_i + \varepsilon_i$$

An instrument must satisfy two conditions:

1. First stage: $Cov(S_i, Z_i) \neq 0$
   - I.e., instrument is correlated with treatment (how to test?)

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Consider an instrument Z_i for S_i in the causal model

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An instrument must satisfy two conditions:

1. First stage: $Cov(S_i, Z_i) \neq 0$
   - I.e., instrument is correlated with treatment (how to test?)
2. Exclusion restriction: $Cov(\varepsilon_i, Z_i) = 0$
   - Can think of as two sub-conditions:
     - Z_i is uncorrelated with unobserved determinants of earnings
     - Z_i has no direct effect on Y_i
   - Exclusion is not testable (caveats later)

The IV coefficient

The exclusion restriction implies

$$Cov(Y_i, Z_i) = Cov(\alpha + \rho S_i + \varepsilon_i, Z_i) = \rho Cov(S_i, Z_i)$$

If the first stage holds, we can divide by $Cov(S_i, Z_i)$:

$$\frac{Cov(Y_i, Z_i)}{Cov(S_i, Z_i)} = \rho$$

- This ratio of covariances is the population IV coefficient

The reduced form and the first stage

Divide the numerator and denominator by $Var(Z_i)$:

$$\frac{Cov(Y_i, Z_i) / Var(Z_i)}{Cov(S_i, Z_i) / Var(Z_i)} = \rho$$

What is the interpretation of this expression?

The reduced form and the first stage

Divide the numerator and denominator by $Var(Z_i)$:

$$\frac{Cov(Y_i, Z_i) / Var(Z_i)}{Cov(S_i, Z_i) / Var(Z_i)} = \rho$$

What is the interpretation of this expression?

- It is a ratio of two OLS regression coefficients:

$$\text{Reduced form:} \quad Y_i = \lambda + \delta Z_i + u_i$$
$$\text{First stage:} \quad S_i = \kappa + \pi Z_i + \eta_i$$

- $\rho_{IV} = \delta / \pi$, the reduced form divided by the first stage

If both IV assumptions hold, then $\rho_{IV} = \rho$

IV example: Angrist and Krueger (1991)

Where do instruments come from?

- Good instruments usually combine:
  - Institutional knowledge
  - Plausible assumptions about behavioral relationships

Classic example: Angrist and Krueger (1991) on the returns to education

Angrist and Krueger (1991)

AK estimate the causal effect of schooling on earnings

Their IV strategy is motivated by the interaction between compulsory schooling and age-at-entry laws

- Students can typically drop out of school on the day they turn 16
- Birth date cutoff for starting age: kids usually start school in the fall of the calendar year they turn six

This generates differences in mean educational attainment by date of birth:

Angrist and Krueger (1991)

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- Birth date cutoff for starting age: kids usually start school in the fall of the calendar year they turn six

This generates differences in mean educational attainment by date of birth:

Birth date	School start date	Dropout date	Schooling at dropout date
January 2, 1930	September 1, 1936	January 2, 1946	9.5 years
December 31, 1930	September 1, 1936	December 31, 1946	10.5 years

Angrist and Krueger (1991)

Dropout + age-at-entry rules $\rightarrow$ less compulsory schooling if born early in the year

- Suggests date of birth as an instrument for schooling
- AK use a "quarter of birth" instrument, an indicator for being born early in the year (January–March)

What are the first stage and exclusion restriction assumptions here?

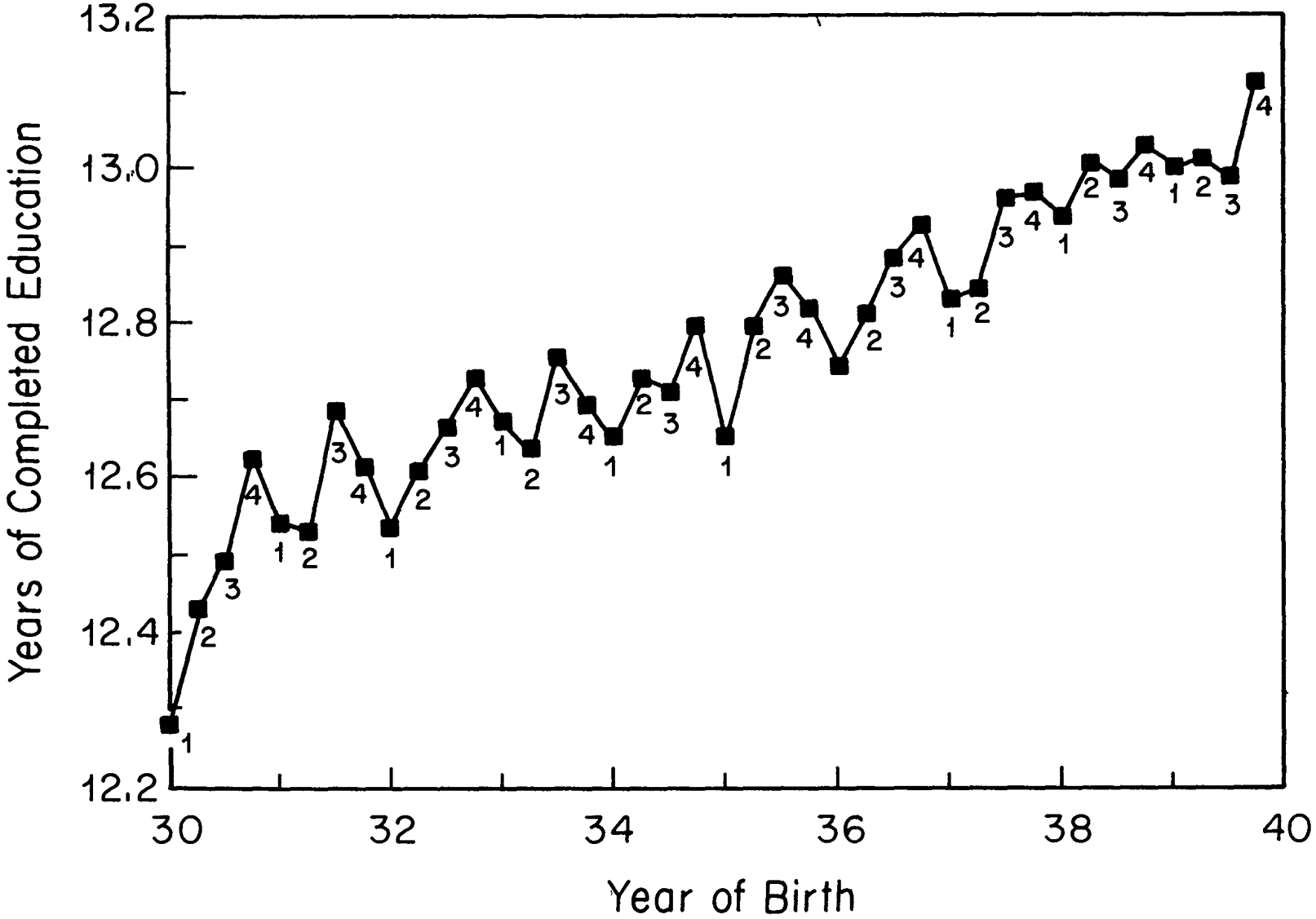


FIGURE I
Years of Education and Season of Birth
1980 Census
Note. Quarter of birth is listed below each observation.

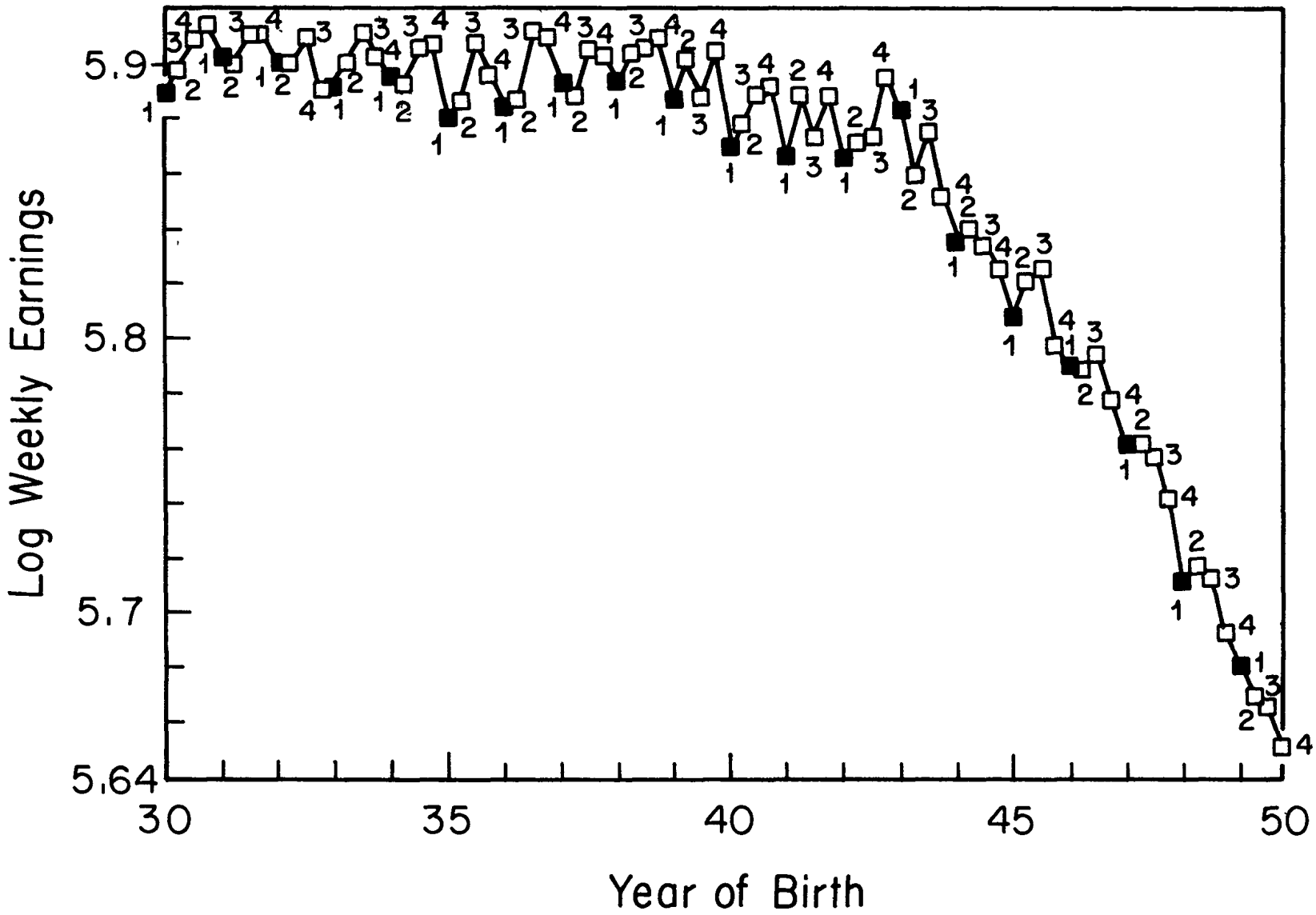


FIGURE V
Mean Log Weekly Wage, by Quarter of Birth
All Men Born 1930–1949; 1980 Census

The Wald estimator

Recall the IV coefficient:

$$\rho_{IV} = \frac{Cov(Y_i, Z_i) / Var(Z_i)}{Cov(S_i, Z_i) / Var(Z_i)}$$

When Z_i is binary, the FS and RF become differences in means:

$$\rho_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[S_i|Z_i = 1] - E[S_i|Z_i = 0]}$$

- This is the population analogue of the Wald estimator

Panel B: Wald Estimates for 1980 Census—Men Born 1930–1939

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) − (2)
ln (wkly. wage)	5.8916	5.9027	−0.01110
			(0.00274)
Education	12.6881	12.7969	−0.1088
			(0.0132)
Wald est. of return to education			0.1020
			(0.0239)
OLS return to education			0.0709
			(0.0003)

IV with covariates

When doing IV it's often necessary to control for additional variables

- E.g. QOB is correlated with age, which likely also affects earnings
- This suggests a violation of exclusion, so we may want to control for smooth functions of age

The long regression of interest now includes the control vector X_i:

$$Y_i = \alpha + \rho S_i + X_i'\gamma + v_i$$

- X_i includes an age polynomial (plus possibly other variables)
- Control variables that are neither instruments nor instrumented are "exogenous covariates"

IV with covariates

Reduced form and first stage regressions augmented with covariates:

$$Y_i = \lambda + \delta Z_i + X_i'\psi + u_i$$
$$S_i = \kappa + \pi Z_i + X_i'\phi + \eta_i$$

As before we can solve for ρ as δ/π; applying the Frisch-Waugh-Lovell theorem, this equals

$$\frac{\delta}{\pi} = \frac{Cov\left(Y_i, \tilde{Z}_i\right)}{Cov\left(S_i, \tilde{Z}_i\right)}$$

Indirect least squares

$$\frac{\delta}{\pi} = \frac{Cov\left(Y_i, \tilde{Z}_i\right)}{Cov\left(S_i, \tilde{Z}_i\right)}$$

- The sample analogue of this ratio is called the indirect least squares (ILS) estimator
- Controlling for covariates is equivalent to using the residual from a regression of Z_i on X_i as the instrument
- The first stage and exclusion restriction assumptions now need to hold for $\tilde{Z}_i$ rather than Z_i
- Importantly, the first stage relationship between S_i and Z_i must survive the addition of the controls X_i

Multiple instruments

Suppose we have instruments $Z_{i1}, \ldots, Z_{iJ}$, collected in the $J \times 1$ vector Z_i

- E.g. dummies for each quarter of birth, rather than just the first one

It's no longer clear how to solve for the long regression coefficient ρ: we could use any one of the J instruments, or any linear combination of them

- ρ is now overidentified: there are multiple ways to solve for the parameter of interest

How should we combine the available information?

Multiple instruments

Our structural equation of interest is still

$$Y_i = \alpha + \rho S_i + X_i'\gamma + \varepsilon_i$$

Consider a first-stage regression of S_i on all the instruments and exogenous controls:

$$S_i = \kappa + Z_i'\pi + X_i'\phi + \eta_i$$

Multiple instruments

Substitute the first stage regression into the structural equation:

$$\begin{aligned} Y_i &= \alpha + \rho\left[\kappa + Z_i'\pi + X_i'\phi + \eta_i\right] + X_i'\gamma + \varepsilon_i \\ &= \alpha + \rho\left(\kappa + Z_i'\pi + X_i'\phi\right) + X_i'\gamma + (\rho\eta_i + \varepsilon_i) \\ &= \alpha + \rho\hat{S}_i + X_i'\gamma + \nu_i \end{aligned}$$

- $\hat{S}_i = \kappa + Z_i'\pi + X_i'\phi$ is the predicted value of S_i from the first stage
- The new residual is $\nu_i = \rho\eta_i + \varepsilon_i = \rho(S_i - \hat{S}_i) + \varepsilon_i$

Two-stage least squares

$$Y_i = \alpha + \rho \hat{S}_i + X_i' \gamma + \nu_i$$

$\hat{S}_i$ and X_i are uncorrelated with ν_i (why?)

- $\rightarrow$ OLS recovers the parameters of this equation

This is the two-stage least squares (2SLS) estimator:

1. Regress S_i on all the instruments and exogenous controls, and obtain predicted values $\hat{S}_i$
2. Regress Y_i on $\hat{S}_i$ and the exogenous controls

2SLS provides one method for combining instruments. Do you think it is the most efficient?

IV as GMM

With N observations on (Y_i, S_i, Z_i, X_i), write the structural equation as

$$Y_i = \alpha + S_i'\rho + X_i'\gamma + \varepsilon_i$$

- S_i is a $K \times 1$ vector of endogenous variables
- Z_i is a $J \times 1$ vector of instruments ($J \geq K$)
- X_i is an $L \times 1$ vector of exogenous controls
- $\mathbf{Z}_i = (1, X_i', Z_i')'$ collects instruments and controls

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We know (from economic theory or institutional knowledge) that

$$\mathrm{E}[\mathbf{Z}_i\varepsilon_i] = \mathrm{E}[\mathbf{Z}_i(y_i - \alpha - S_i'\rho - X_i'\gamma)] = \mathrm{E}[m_i(\theta)] = 0$$

where $\theta = (\rho; \gamma)$

What do we call this? How many equations? How many unknowns?

Casting everything as GMM

Recall our class on GMM (Estimation slide deck)

- GMM minimizes the following criterion function:

$$Q_{GMM}(\theta) = \mathrm{E}[m_i(\theta)' \mathbf{W} m_i(\theta)] \tag{1}$$

- What happens to $\mathbf{W}$ when $J = K$?

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- How do we choose $\mathbf{W}$ when $J > K$?
- Efficient GMM: $\mathbf{W}^* = \mathbf{\Omega}^{-1}$, where $\mathbf{\Omega} = \mathrm{E}[m_i(\theta) m_i(\theta)'] = \mathrm{E}[\varepsilon_i^2 \mathbf{Z}_i \mathbf{Z}_i']$
  - Under homoskedasticity, $\mathrm{E}[\varepsilon_i^2 | \mathbf{Z}_i] = \sigma^2$, so $\mathbf{\Omega}^{-1} = \sigma^{-2} \mathrm{E}[\mathbf{Z}_i \mathbf{Z}_i']^{-1}$
  - With heteroskedasticity, $\mathbf{\Omega}$ must be estimated: $\hat{\mathbf{\Omega}} = \frac{1}{N} \sum_i \hat{\varepsilon}_i^2 \mathbf{Z}_i \mathbf{Z}_i'$ using 2SLS residuals (two-step GMM)

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- Turns out that 2SLS minimizes (1) by setting $\mathbf{W} = E[\mathbf{Z}_i \mathbf{Z}_i']^{-1}$
  - (See MHE 4.2.1 for the proof)
  - What does this mean for efficiency of 2SLS?

Over-id test

The GMM formulation is also useful for testing the validity of the model

- When there are more instruments than endogenous variables ($J > K$), there are multiple ways to solve for the structural parameters

An over-id test (see Estimation slides) tells us how close the moment conditions are to being satisfied

- In the IV setup, an overid test checks whether all the IV moment restrictions are close enough to being satisfied

‘’

Overid tests in the IV world

Let $\hat{\theta}$ be the minimizer of the sample GMM criterion function:

$$\hat{\theta} = \underset{c}{\operatorname{argmin}}\, \hat{Q}_{GMM}(c) = N\hat{m}_i(c)'\mathbf{W}\hat{m}_i(c)$$

where $\hat{m}_i(\theta) = \frac{1}{N}\mathbf{Z}_i(y_i - \alpha - S_i'\rho - X_i'\gamma)$ (evaluated at the "truth" θ)

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If $\mathbf{W} = \mathrm{E}[\sqrt{N}\hat{m}_i(\theta)\sqrt{N}\hat{m}_i(\theta)']^{-1}$, then as $N \to \infty$

$$\hat{Q}_{GMM}(\hat{\theta}) \sim \chi^2_{J-K}$$

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If $\mathbf{W} = \mathrm{E}[\sqrt{N}\hat{m}_i(\theta)\sqrt{N}\hat{m}_i(\theta)']^{-1}$, then as $N \to \infty$

$$\hat{Q}_{GMM}(\hat{\theta}) \sim \chi^2_{J-K}$$

- If $\hat{Q}_{GMM}(\hat{\theta})$ is large relative to the critical value of χ^2_{J-K}, the model is rejected
- Rejection means that one of the J instruments is false: there is one instrument for which $\mathrm{E}[Z_i\varepsilon_i]$ is actually not zero
- Sometimes this is also called the J-test in the IV setup

Intuition

Suppose there are no controls and we have one endogenous variable S_i and two potential instruments Z_{i1} and Z_{i2}

Two instruments give us two potential (simple) IV estimates of ρ

- First estimate of ρ obtained by solving the moment condition

$$\mathrm{E}[Z_{i1}(y_i - S_i\rho_1)] = 0$$

- And the second is obtained by fitting

$$\mathrm{E}[Z_{i2}(y_i - S_i\rho_2)] = 0$$

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Over-id tests whether a model that imposes $\rho_2 = \rho_1$ can fit the data

- What is the key underlying assumption that makes this test useful?

2SLS mistakes

2SLS is the workhorse IV estimator used by empirical researchers

It's worth covering some common pitfalls related to 2SLS estimation

2SLS mistakes 1: standard errors

It's usually not a good idea to run the two stages of 2SLS manually

- For one thing, this will give the wrong standard errors

Consider a setting with one endogenous variable S_i:

$$Y_i = \alpha + \rho S_i + \varepsilon_i$$

If you run 2SLS manually, the second stage is

$$Y_i = \alpha + \rho \hat{S}_i + v_i$$

where $v_i = \varepsilon_i + \rho(S_i - \hat{S}_i)$

Manual 2SLS will typically generate standard errors that are too large

- The residual variance should be calculated from the structural residual $Y_i - \hat{\alpha} - \hat{\rho} S_i$, not the second-stage residual $Y_i - \hat{\alpha} - \hat{\rho} \hat{S}_i$

2SLS mistakes 2: covariate ambivalence

A second possible mistake in manual 2SLS is the treatment of covariates

Suppose we run the 2SLS procedure

$$S_i = \kappa + Z_i'\pi + X_{1i}'\phi_1 + \eta_i$$
$$Y_i = \alpha + \rho\hat{S}_i + X_{1i}'\theta_1 + X_{2i}'\theta_2 + v_i$$

- The covariate vector X_{2i} is included in the second stage but not the first (the opposite of the instruments Z_i)

What's wrong with this?

2SLS mistakes 2: covariate ambivalence

$$S_i = \kappa + Z_i'\pi + X_{1i}'\phi_1 + \eta_i$$
$$Y_i = \alpha + \rho\hat{S}_i + X_{1i}'\theta_1 + X_{2i}'\theta_2 + v_i$$

- The second-stage residual is $v_i = \varepsilon_i + \rho(S_i - \hat{S}_i)$
- Since $S_i - \hat{S}_i$ is a residual from a regression that includes X_{1i}, $Cov(X_{1i}, S_i - \hat{S}_i) = 0$
- But in general $Cov(S_i - \hat{S}_i, X_{2i}) \neq 0$, so we may have created a correlation between the error term and covariates, leading to bias in all parameter estimates
- "If a covariate is good enough for the second stage, it's good enough for the first"

2SLS mistakes 3: the forbidden regression

In response to repeated mistakes by graduate students, Jerry Hausman labeled one 2SLS mistake "the forbidden regression"

The forbidden regression involves the erroneous application of 2SLS logic to nonlinear models

Suppose there is a dummy endogenous variable $T_i \in \{0, 1\}$, a vector of covariates X_i, and a vector of instruments Z_i

The causal equation is

$$Y_i = \alpha + \rho T_i + X_i'\theta + \varepsilon_i$$

2SLS mistakes 3: the forbidden regression

First stage regression:

$$T_i = \kappa + Z_i'\pi + X_i'\phi + \eta_i$$

Since T_i is binary, $E\left[T_i|Z_i, X_i\right]$ is likely nonlinear unless the model is saturated, so the first stage is a misspecified model of the CEF

One might be tempted to try a nonlinear model to better approximate the CEF, e.g. a probit:

$$Pr\left[T_i = 1|X_i, Z_i\right] = \Phi\left(\kappa + Z_i'\pi + X_i'\phi\right)$$

2SLS mistakes 3: the forbidden regression

The nonlinear first stage can be used to obtain predicted values:

$$\hat{T}_i = \Phi\left(\hat{\kappa} + Z_i'\hat{\pi} + X_i'\hat{\phi}\right)$$

One might then be tempted to run a 2SLS-style second stage:

$$Y_i = \alpha + \rho\,\hat{T}_i + X_i'\theta + \varepsilon_i$$

This regression is forbidden!

2SLS mistakes 3: the forbidden regression

To see why, rewrite the structural equation:

$$Y_i = \alpha + \rho \hat{T}_i + X_i'\theta + \left(\varepsilon_i + \rho(T_i - \hat{T}_i)\right)$$

- If the first stage is estimated via OLS, the residual $(T_i - \hat{T}_i)$ is guaranteed to be uncorrelated with the fit $\hat{T}_i$ and with the included covariates X_i
- Nonlinear residuals do not have this property: $Cov\left(T_i - \hat{T}_i, \hat{T}_i\right)$ and $Cov\left(T_i - \hat{T}_i, X_i\right)$ will generally equal zero only when $E\left[T_i | X_i, Z_i\right]$ is really a probit
- Who knows whether the probit CEF is right? Better to stick with a linear first stage, which guarantees zero covariance between residuals and fits/regressors even if the CEF is misspecified

Pop-quiz

Suppose you came up with what you think is a good instrument, Z_i, but you're not 100% sure about the exclusion restriction

Inspired by your knowledge of regression and the meaning of the exclusion restriction, you come up with the following test for the validity of the instrument:

$$y_i = \alpha + \beta S_i + \gamma Z_i + r_i$$

That is, you run a regression of your outcome on the endogenous variable S_i and the instrument Z_i

Since the instrument only operates through S_i, you argue that if the exclusion restriction is valid, then $\beta = \rho$ (the true causal effect) and $\gamma = 0$

Right? Wrong?

Many/weak instruments: the problem

IV is biased in finite samples, since it is a ratio of reduced form and first stage:

$$E\left[\hat{\rho}_{IV}\right] = E\left[\frac{\hat{\delta}}{\hat{\pi}}\right] \neq \frac{E[\hat{\delta}]}{E[\hat{\pi}]} = \rho$$

- Just-identified IV: a ratio of two (jointly normal) estimates $\rightarrow$ no moments and fat tails (Cauchy as $\pi \rightarrow 0$), but approximately median unbiased: $Q_{0.5}(\hat{\rho}_{IV}) \approx \rho$
- Overidentified 2SLS is worse: it is centered in the wrong place, biased towards OLS
- The bias grows with the number of instruments (overfitting the first stage) and as the first stage gets weaker $\rightarrow$ "many/weak" instruments
- Bound, Jaeger and Baker (1995) showed this matters in practice

Bound, Jaeger and Baker (1995)

BJB revisit the Angrist and Krueger QOB instrument

- The Wald estimates we studied are just-identified, so the many/weak critique does not apply to them
- But AK also report 2SLS estimates that add QOB $\times$ year-of-birth (30) and QOB $\times$ state-of-birth (150) interactions as instruments, overidentified by 179
  - Rationale: compulsory schooling laws differ across states and cohorts
  - Asymptotically this can only help: better first-stage fit, smaller standard errors

The QOB first stage is small and most interactions are irrelevant, so this is a severe many/weak scenario

BJB's experiment: replace QOB with randomly generated fake QOB, which has no first stage by construction, and re-run AK's 2SLS

Bound, Jaeger and Baker (1995)

	2SLS (real QOB)	2SLS (fake QOB)	OLS
30 QOB*year	0.081	0.062	0.063
interactions	(0.016)	(0.037)	(0.000)
180 QOB*year and state	0.083	0.060	0.063
interactions	(0.009)	(0.015)	(0.000)

- With junk instruments, 2SLS essentially reproduces OLS
- Conventional standard errors give no warning: the fake-QOB, 180-instrument estimate looks precise despite having no first stage
- Where does this bias come from, and how do we diagnose and correct it?

Where the bias comes from: setup

One endogenous regressor, Q instruments, in matrix form:

$$\begin{aligned} \text{Second stage:} \quad & Y = \rho S + \varepsilon \\ \text{First stage:} \quad & S = Z\pi + \eta \end{aligned}$$

2SLS regresses Y on $\hat{S} = P_Z S$, so

$$\hat{\rho}_{2SLS} - \rho = (S' P_Z S)^{-1} S' P_Z \varepsilon = (S' P_Z S)^{-1} \left[\pi' Z' \varepsilon + \eta' P_Z \varepsilon\right]$$

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One endogenous regressor, Q instruments, in matrix form:

$$\begin{aligned} \text{Second stage:} \quad & Y = \rho S + \varepsilon \\ \text{First stage:} \quad & S = Z\pi + \eta \end{aligned}$$

2SLS regresses Y on $\hat{S} = P_Z S$, so

$$\hat{\rho}_{2SLS} - \rho = (S' P_Z S)^{-1} S' P_Z \varepsilon = (S' P_Z S)^{-1} \left[\pi' Z' \varepsilon + \eta' P_Z \varepsilon\right]$$

We want the expectation of this. Problem: $(S' P_Z S)^{-1}$ is random and correlated with the bracket (why?)

Bekker (1994): under asymptotics where Q grows with N, replacing $(S' P_Z S)^{-1}$ by $(E[S' P_Z S])^{-1}$ is a good approximation:

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \left(E[S' P_Z S]\right)^{-1} E\left[\pi' Z' \varepsilon + \eta' P_Z \varepsilon\right]$$

Where the bias comes from: the trace trick

Condition on Z and take the numerator one term at a time:

- First term: $E[\pi' Z' \varepsilon \mid Z] = \pi' Z' E[\varepsilon \mid Z] = 0$ (why?)

Where the bias comes from: the trace trick

Condition on Z and take the numerator one term at a time:

- First term: $E[\pi' Z' \varepsilon \mid Z] = \pi' Z' E[\varepsilon \mid Z] = 0$ (why?)
- Second term:

$$\begin{aligned} E[\eta' P_Z \varepsilon \mid Z] &= E\left[\operatorname{tr}(\eta' P_Z \varepsilon) \mid Z\right] = E\left[\operatorname{tr}(P_Z \varepsilon \eta') \mid Z\right] \\ &= \operatorname{tr}\left(P_Z \, E[\varepsilon \eta' \mid Z]\right) = \sigma_{\eta\varepsilon} \operatorname{tr}(P_Z) = \sigma_{\eta\varepsilon} \, Q \end{aligned}$$

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- Uses two tricks: a scalar equals its trace, the trace is cyclic
- (also relies on IID residual, can be adapted)
- Nonzero because η and ε are correlated (why?)
  - Intuition: each instrument lets $\hat{S}$ pick up a bit of η, hence of ε. More instruments, more leakage

The many/weak bias formula

Same trick on the denominator:
$E[S'P_Z S] = E[\pi' Z' Z \pi] + E[\eta' P_Z \eta] = E[\pi' Z' Z \pi] + \sigma_\eta^2 \, Q$. Hence

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\varepsilon} \, Q}{E[\pi' Z' Z \pi] + \sigma_\eta^2 \, Q} = \frac{\sigma_{\eta\varepsilon}}{\sigma_\eta^2} \cdot \frac{1}{1+F}, \qquad F \equiv \frac{E[\pi' Z' Z \pi]}{\sigma_\eta^2 \, Q}$$

where F is the population first-stage F-statistic for the joint significance of the instruments

Comparing to OLS

Let $V \equiv Var(Z_i'\pi)$. OLS regresses Y on S, and $Var(S_i) = V + \sigma_\eta^2$, so

$$\text{OLS bias} = \frac{Cov(S_i, \varepsilon_i)}{Var(S_i)} = \frac{\sigma_{\eta\varepsilon}}{V + \sigma_\eta^2}$$

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$$\text{OLS bias} = \frac{Cov(S_i, \varepsilon_i)}{Var(S_i)} = \frac{\sigma_{\eta\varepsilon}}{V + \sigma_\eta^2}$$

Rewrite what we just derived in these terms:

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\varepsilon}}{\sigma_\eta^2} \cdot \frac{1}{1+F} = \frac{\sigma_{\eta\varepsilon}}{V + \sigma_\eta^2} \cdot \frac{1 + V/\sigma_\eta^2}{1+F} \approx \underbrace{\frac{\sigma_{\eta\varepsilon}}{V + \sigma_\eta^2}}_{\text{OLS bias}} \cdot \frac{1}{1+F}$$

(what gives us the last step?)

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(what gives us the last step?)

Nice way to think about IV bias!

- When F is small, 2SLS is more biased towards OLS
- $F = E[\pi' Z' Z \pi]/(\sigma_\eta^2 Q)$ small when instruments weak (small π) or many (large Q)

A different way to compare bias

With centered instruments $E[\pi' Z' Z \pi] = N\,V$, so the same expression is

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\varepsilon}\,Q}{N\,V + Q\,\sigma_\eta^2} = \underbrace{\frac{\sigma_{\eta\varepsilon}}{V + \sigma_\eta^2}}_{\text{OLS bias}} \times \frac{Q\,(V + \sigma_\eta^2)}{N\,V + Q\,\sigma_\eta^2}$$

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The second factor equals 1 in two cases:

- No first stage, $V = 0$
- As many instruments as observations, $Q = N$

Many instruments (high Q/N) with little signal (low V/σ_η^2) is the bad case

Many/weak in practice

Diagnose: report the first-stage F on the instruments

- Staiger and Stock (1997) rule of thumb: $F > 10$
- Lee, McCrary, Moreira and Porter (2022): screening on $F > 10$ causes inference issues (pre-testing); need a larger F (104!) for correct size in just-id IV
- Provide the tF correction for smaller F

Weak-IV robust inference: Anderson-Rubin is valid whatever the instrument strength

- Survey: Andrews, Stock and Sun (2019)

JIVE (Angrist, Imbens and Krueger 1999)

The bias came from i's own first-stage error entering i's fitted value, so estimate the first stage leaving i out and use $\hat{S}_{-i} = Z_i' \hat{\pi}_{-i}$ as the instrument: jackknife IV

- (Read the NBER WP, not the JAE)

Why it helps: $\hat{S}_{-i}$ does not depend on η_i, so $E[\eta_i \hat{S}_{-i} \mid Z] = 0$ and the $\eta' P_Z \varepsilon$ term disappears. JIVE is consistent under many-instrument asymptotics and with heteroskedasticity

Caveat: conventional 2SLS-style standard errors for JIVE are wrong in many/weak settings

JIVE does not solve all our problems

Ackerberg and Devereux (2009): leaving out i kills the own-observation term, but the JIVE estimators of Angrist, Imbens and Krueger (1999) remain substantially biased with many/weak instruments

- The culprit is the exogenous covariates: JIVE jackknifes them together with the instruments, which reintroduces a bias term of order L/N (L = number of covariates)
- Fix: partial the covariates out first, then jackknife (IJIVE); a further Nagar-type correction removes the remaining higher-order bias (UIJIVE)
- Monte Carlo: JIVE has large bias and dispersion when instruments are weak; IJIVE/UIJIVE behave like LIML, and unlike LIML stay consistent under heteroskedasticity

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Bottom line: leave-out fixes the bias from many instruments; it does nothing about weak identification, which still shows up as bias, dispersion, and invalid standard errors

UJIVE (Kolesár 2013)

Many-instrument designs usually also have many covariates W (e.g. fixed effects for randomization cells)

With many covariates, JIVE is inconsistent under many-instrument asymptotics: the covariates must be jackknifed too

UJIVE: leave i out of both the full first stage (on Z and W) and the covariate-only regression (on W), and instrument with the difference,

$$\tilde{S}_i = \frac{(P_{ZW}S)_i - (P_{ZW})_{ii}S_i}{1-(P_{ZW})_{ii}} - \frac{(P_W S)_i - (P_W)_{ii}S_i}{1-(P_W)_{ii}}$$

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Why it matters beyond bias: with heterogeneous effects, LIML and other k-class estimators are inconsistent for any causal parameter under many instruments. UJIVE is consistent for a convex combination of LATEs, and is robust to heteroskedasticity

In practice: use UJIVE rather than a leave-out instrument with covariates bolted on, and report many-instrument standard errors

The judges design

Cases are randomly assigned (within court $\times$ year cells) to judges who differ in leniency. Examples: Kling (2006) on sentencing, Doyle (2007) on foster care, Dobbie (2015) on bankruptcy

Instrument T_i with the leave-out leniency of i's judge $j(i)$, which is JIVE on a full set of judge dummies:

$$T_i^* = \frac{1}{n_{j(i)} - 1} \sum_{k \neq i} 1\{j(k) = j(i)\} T_k$$

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Question: what features of this design make many/weak problems likely?

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Question: what features of this design make many/weak problems likely?

- Many weak instruments: hundreds of judges, each only slightly different from average
- Many covariates: the cells that deliver random assignment. These leak even when you JIVE $\rightarrow$ UJIVE

Many/weak inference: summary

Just-identified, one instrument. Report the first stage. Conventional t-test needs $F > 104$ (Lee et al. 2022); otherwise use their tF critical values or the Anderson-Rubin (AR) test, which is valid for any F. Angrist and Kolesár (2024): with modest endogeneity the usual inference is fine even at low F

Overidentified, few instruments. 2SLS is biased towards OLS. Diagnose with the partial F and Stock-Yogo (2005) critical values; with heteroskedasticity or clustering use the effective F of Montiel Olea and Pflueger (2013). Weak-IV robust confidence sets: AR or conditional likelihood ratio (Moreira 2003). LIML/Fuller for approximate median unbiasedness

Many instruments (judges, examiners). 2SLS is unreliable. Use UJIVE or HFUL (Hausman et al. 2012). Standard errors need the many-instrument correction (Chao et al. 2012; Kolesár 2013). If identification is also weak, the JIVE t-test fails: use the jackknife AR test and pre-test for identification strength (Mikusheva and Sun 2022)

Always: show the first stage and the reduced form. When in doubt, AR

Section 2

Part II: Heterogeneous Treatment Effects

Heterogeneous effects

Our discussion of IV has focused on models with constant causal effects; we next consider models with heterogeneous effects

- Understanding effect heterogeneity is important for assessing external validity: when does a causal effect have predictive value for other populations or time periods?
- Effect heterogeneity also gives econometric models economic content (e.g. self-selection and comparative advantage)

Heterogeneous effects: setup

Begin with a binary treatment, binary instrument model:

- Outcome Y_i
- Treatment $T_i \in \{0, 1\}$
- Instrument $Z_i \in \{0, 1\}$

Define potential outcomes $Y_i(t, z)$, now indexed against both T_i and Z_i

Define potential treatment statuses, $T_i(1)$ and $T_i(0)$

- $T_i(z)$ describes how individual i responds to the instrument

Observe Z_i, $T_i = T_i(Z_i)$, and $Y_i = Y_i(T_i, Z_i)$

Assumption 1: independence

First assumption in the heterogeneous effects framework: independence

The instrument is assumed independent of potential outcomes and potential treatment statuses:

$$(Y_i(1,1), Y_i(1,0), Y_i(0,1), Y_i(0,0), T_i(1), T_i(0)) \perp Z_i$$

- This means the instrument is as good as randomly assigned
- Independence implies that regressions of Y_i and T_i on Z_i identify causal effects of the instrument

Assumption 2: exclusion

Second assumption: exclusion of the instrument,

$$Y_i(t,1) = Y_i(t,0) \quad \forall (i,t)$$

- The instrument has no direct causal effect on the outcome: Z_i can only shift Y_i by shifting T_i
- Random assignment does not give you exclusion; you must specify the right causal channel
- Exclusion allows us to write potential outcomes as $Y_i(t)$. Models that begin with this have implicitly imposed exclusion
- Exclusion and independence are distinct, though they are often grouped together in the single restriction

$$(Y_i(1), Y_i(0), T_i(1), T_i(0)) \perp Z_i$$

Assumption 3: first stage

Third assumption: the existence of a first stage,

$$Pr\left[T_i = 1 | Z_i = 1\right] > Pr\left[T_i = 1 | Z_i = 0\right]$$

- The probability of treatment with the instrument switched on must exceed this probability with the instrument switched off
- This assumption is testable
- The direction of the inequality is not important; the key is that the instrument shifts the probability of treatment

Assumption 4: monotonicity

Fourth assumption: monotonicity,

$$T_i(1) \geq T_i(0) \quad \forall i$$

- Monotonicity requires that the instrument only moves treatment status in one direction
- Z_i either induces an individual to take the treatment, or does nothing
- Relative to constant effects IV, monotonicity is the novel assumption

A popular IV

Draft lottery (1969)

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On December 1, 1969, the Selective Service System of the United States conducted two lotteries to determine the order of call to military service in the Vietnam War in the year 1970, for men born from January 1, 1944 to December 31, 1950. These lotteries occurred during a period of conscription in the United States that lasted from 1947 to 1973. It was the first time a lottery system had been used to select men for military service since 1942. The lottery would establish the priority of call based on the birth dates of registrants.

Subpopulations

Monotonicity implies that we can partition the population into three groups:

1. Never takers have $T_i(1) = T_i(0) = 0$: they don't take the treatment irrespective of the instrument
2. Always takers have $T_i(1) = T_i(0) = 1$: they take the treatment even if assigned $Z_i = 0$
3. Compliers have $T_i(1) = 1$ and $T_i(0) = 0$ (equivalently, $T_i(1) > T_i(0)$): they take the treatment if assigned $Z_i = 1$, but not if assigned $Z_i = 0$

Monotonicity rules out the existence of defiers, individuals with $T_i(1) = 0$ but $T_i(0) = 1$

This terminology comes from randomized trials where Z_i is assignment to treatment and T_i is receipt of treatment

The LATE framework

1. Independence/exclusion: $(Y_i(1), Y_i(0), T_i(1), T_i(0)) \perp Z_i$
2. First stage: $Pr\left[T_i = 1 | Z_i = 1\right] > Pr\left[T_i = 1 | Z_i = 0\right]$
3. Monotonicity: $T_i(1) \geq T_i(0)\ \forall i$

These three (really, four) assumptions constitute the Local Average Treatment Effect (LATE) framework (Imbens and Angrist 1994; Angrist, Imbens and Rubin 1996)

The LATE theorem

In the LATE framework, IV estimates can be interpreted as average causal effects for compliers, known as Local Average Treatment Effects:

$$\frac{E\left[Y_i|Z_i=1\right]-E\left[Y_i|Z_i=0\right]}{E\left[T_i|Z_i=1\right]-E\left[T_i|Z_i=0\right]}=E\left[Y_i(1)-Y_i(0)|\,T_i(1)>T_i(0)\right]\equiv LATE$$

- The left-hand side is the IV (Wald) estimand
- The right-hand side is the average effect of treatment for compliers, individuals who respond to the instrument by changing treatment status
- This effect is "local" in the sense that it is specific to the instrument used: different instruments may yield different LATEs

The LATE theorem: proof

The numerator of the Wald estimand, the reduced form, is

$$\begin{aligned}
E\left[Y_i|Z_i=1\right]-E\left[Y_i|Z_i=0\right] &= E\left[Y_i(T_i(1))|Z_i=1\right]-E\left[Y_i(T_i(0))|Z_i=0\right] \\
&= E\left[Y_i(T_i(1))-Y_i(T_i(0))\right] \\
&= E\left[Y_i(T_i(1))-Y_i(T_i(0))|\,T_i(1)\neq T_i(0)\right]Pr\left[T_i(1)\neq T_i(0)\right] \\
&= E\left[Y_i(1)-Y_i(0)|\,T_i(1)>T_i(0)\right]Pr\left[T_i(1)>T_i(0)\right]
\end{aligned}$$

The LATE theorem: proof

The denominator of the Wald estimand, the first stage, is

$$\begin{aligned} E\left[T_i|Z_i=1\right]-E\left[T_i|Z_i=0\right] &= E\left[T_i(1)|Z_i=1\right]-E\left[T_i(0)|Z_i=0\right] \\ &= E\left[T_i(1)-T_i(0)\right] \\ &= Pr\left[T_i(1)>T_i(0)\right] \end{aligned}$$

The LATE theorem: proof

Then the Wald estimand is

$$\frac{E\left[Y_i|Z_i=1\right]-E\left[Y_i|Z_i=0\right]}{E\left[T_i|Z_i=1\right]-E\left[T_i|Z_i=0\right]} = \frac{E\left[Y_i(1)-Y_i(0)|T_i(1)>T_i(0)\right]Pr\left[T_i(1)>T_i(0)\right]}{Pr\left[T_i(1)>T_i(0)\right]}$$
$$= E\left[Y_i(1)-Y_i(0)|T_i(1)>T_i(0)\right]$$

The LATE theorem: discussion

The LATE model is nonparametric: the LATE theorem applies whether Y_i is binary, continuous, count, or any other distribution

The LATE theorem provides useful guidance for interpreting IV estimates and thinking about external validity

Who are the compliers for the AK quarter of birth instrument?

Birth date	School start date	Dropout date	Schooling at dropout date
January 2, 1930	September 1, 1936	January 2, 1946	9.5 years
December 31, 1930	September 1, 1936	December 31, 1946	10.5 years

Panel B: Wald Estimates for 1980 Census—Men Born 1930–1939

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) − (2)
ln (wkly. wage)	5.8916	5.9027	−0.01110
			(0.00274)
Education	12.6881	12.7969	−0.1088
			(0.0132)
Wald est. of return to education			0.1020
			(0.0239)
OLS return to education			0.0709
			(0.0003)

QOB compliers

The QOB compliers are individuals who drop out as soon as possible, and are required to stay longer when their birthdays fall late in the year

Perhaps returns to schooling are higher for these individuals, e.g. because returns are higher at low levels of schooling. Can this explain $IV \geq OLS$?

LATE and other treatment effects

We can now add LATE to our catalogue of treatment effects:

- Average treatment effect: $ATE = E\left[Y_i(1) - Y_i(0)\right]$
- Effect of treatment on the treated: $TOT = E\left[Y_i(1) - Y_i(0) | T_i = 1\right]$
- Effect of treatment on the non-treated: $TNT = E\left[Y_i(1) - Y_i(0) | T_i = 0\right]$
- Local average treatment effect: $LATE = E\left[Y_i(1) - Y_i(0) | T_i(1) > T_i(0)\right]$

How is LATE related to these other, more general parameters?

The Bloom scenario

The $T_i = 1$ population includes

- Compliers with $Z_i = 1$
- Always takers

IV estimates do not capture effects for always takers, so in general $LATE \neq TOT$

Suppose there is one-sided non-compliance, so that $Z_i = 0 \implies T_i = 0$

- Example: a medical trial where researchers can prevent the control group from accessing the treatment but cannot force the treatment group to take it

In this case there are no always takers, so $LATE = TOT$. This is the "Bloom scenario," after Bloom (1984)

LATE example: MDVE

LATE is useful for thinking about randomized experiments where forcing compliance is either impossible or unethical

- Example: the Minneapolis Domestic Violence Experiment (MDVE), an effort to determine the best response to domestic violence calls

Police responding to domestic violence calls can follow two main strategies:

- Arrest the offender (harsh)
- Warning/referral to counseling ("coddling")

Which of these two approaches leads to less domestic violence in the future?

LATE example: MDVE

In the MDVE, police were given randomly ordered color-coded charge sheets, where colors indicated the suggested strategy (arrest or coddle)

Enforcing compliance was unethical: some offenders are extremely violent and may inflict injury if not immediately arrested

- Police were free to ignore suggestions if they deemed it necessary

The original MDVE literature looked only at the reduced-form effect of assignment to arrest ("intent to treat," ITT), but this is a perfect IV scenario

- It's very plausible that suggested arrest would weakly increase the probability of arrest in all cases

LATE example: MDVE

The MDVE maps directly into the IV framework:

- Treatment T_i is arrest
- Instrument Z_i is assignment to arrest
- Outcome Y_i is an indicator for future offense

In practice the police arrested everyone assigned to arrest, and also arrested some offenders assigned to coddling

How should we interpret IV estimates from the MDVE?

LATE example: MDVE

In the MDVE, $Z_i = 1 \implies T_i = 1$

- There are no never takers, since everyone assigned to treatment is treated
- Everyone who is coddled is a complier

This is the flip of the Bloom scenario: $LATE = TNT$

- The MDVE identifies the effect of arrest on the coddled

Characterizing compliers

To interpret any IV estimate, it's useful to ask: who are the compliers? It turns out we can learn a lot about them

Consider the subpopulations captured in each combination of T_i and Z_i:

	$T_i = 1$	$T_i = 0$
$Z_i = 1$	Compliers and always takers	Never takers
$Z_i = 0$	Always takers	Compliers and never takers

Characterizing compliers

	$T_i = 1$	$T_i = 0$
$Z_i = 1$	Compliers and always takers	Never takers
$Z_i = 0$	Always takers	Compliers and never takers

- Always takers are directly observable in the $(Z_i = 0, T_i = 1)$ group
- Never takers are directly observable in the $(Z_i = 1, T_i = 0)$ group
- The groups with $T_i = Z_i$ are mixtures of ATs/compliers and NTs/compliers

But the mixing probabilities are identified, since

$$\pi^{at} = Pr\left[T_i(1) = T_i(0) = 1\right] = Pr\left[T_i = 1 | Z_i = 0\right]$$

$$\pi^{nt} = Pr\left[T_i(1) = T_i(0) = 0\right] = Pr\left[T_i = 0 | Z_i = 1\right]$$

$$\pi^{c} = Pr\left[T_i(1) > T_i(0)\right] = Pr\left[T_i = 1 | Z_i = 1\right] - Pr\left[T_i = 1 | Z_i = 0\right]$$

Characterizing compliers

For an observed covariate X_i we have

$$E[X_i|T_i = Z_i = 1] = \left(\frac{\pi^c}{\pi^c+\pi^{at}}\right) E[X_i|T_i(1) > T_i(0)] + \left(\frac{\pi^{at}}{\pi^c+\pi^{at}}\right) E[X_i|T_i(1) = T_i(0) = 1]$$

- The mean for the $T_i = Z_i = 1$ group is a weighted average of complier and always taker means, with known weights
- The always taker mean is identified since $E[X_i|T_i(1) = T_i(0) = 1] = E[X_i|T_i = 1, Z_i = 0]$
- Then the complier mean is also identified:

$$E[X_i|T_i(1) > T_i(0)] = \left(\frac{\pi^c+\pi^{at}}{\pi^c}\right) E[X_i|T_i = Z_i = 1] - \left(\frac{\pi^{at}}{\pi^c}\right) E[X_i|T_i = 1, Z_i = 0]$$

Characterizing compliers

$$E\left[X_i \middle| T_i(1) > T_i(0)\right] = \left(\frac{\pi^c + \pi^{at}}{\pi^c}\right) E\left[X_i \middle| T_i = Z_i = 1\right] - \left(\frac{\pi^{at}}{\pi^c}\right) E\left[X_i \middle| T_i = 1, Z_i = 0\right]$$

- The complier mean of X_i can be recovered by removing the always taker mean from the complier/always taker mixture, adjusting appropriately for relative probabilities
- This means we can learn the observed characteristics of compliers as if they were observable in our data

Complier mean potential outcomes

We could have substituted the outcome Y_i in place of X_i in this argument

- This implies that the marginal mean of $Y_i(1)$ for compliers, $E\left[Y_i(1)|T_i(1) > T_i(0)\right]$, is also identified
- Likewise, $E\left[Y_i(0)|T_i(1) > T_i(0)\right]$ can be obtained by removing the never taker mean outcome from the $Z_i = T_i = 0$ mixture

The levels of complier mean potential outcomes are therefore identified, not just the difference between them

Computing complier characteristics

Abadie (2002) provides a convenient method for computing complier characteristics

For any measurable function $g(X_i, Y_i)$:

$$\frac{E\left[g(X_i, Y_i)\, T_i | Z_i = 1\right] - E\left[g(X_i, Y_i)\, T_i | Z_i = 0\right]}{E\left[T_i | Z_i = 1\right] - E\left[T_i | Z_i = 0\right]} = E\left[g(X_i, Y_i(1)) | \, T_i(1) > T_i(0)\right]$$

$$\frac{E\left[g(X_i, Y_i)(1 - T_i) | Z_i = 1\right] - E\left[g(X_i, Y_i)(1 - T_i) | Z_i = 0\right]}{E\left[1 - T_i | Z_i = 1\right] - E\left[1 - T_i | Z_i = 0\right]} = E\left[g(X_i, Y_i(0)) | \, T_i(1) > T_i(0)\right]$$

Computing complier characteristics

Abadie's result shows that in the IV system

$$g(X_i, Y_i)\, T_i = \alpha + \gamma\, T_i + u_i$$
$$T_i = \lambda + \pi Z_i + \eta_i$$

the IV coefficient γ recovers the treated complier mean of g:

$$\gamma = E\left[g(X_i, Y_i(1)) \middle| T_i(1) > T_i(0)\right]$$

- Substitute $(1 - T_i)$ for T_i in both equations to obtain the untreated complier mean

Computing complier characteristics

Useful special cases:

- $g(X_i, Y_i) = X_i$: complier mean of X_i
- $g(X_i, Y_i) = Y_i$: marginal complier means of $Y_i(1)$ and $Y_i(0)$
- $g(X_i, Y_i) = 1\{Y_i \leq y\}$: complier CDFs of $Y_i(1)$ and $Y_i(0)$ evaluated at y

This last result shows that the full marginal distributions of $Y_i(1)$ and $Y_i(0)$ are identified for compliers

Angrist and Evans (1998)

- Research question: what is the effect of fertility on the labor supply of women?
- Regressions of Employment on # of Children likely to be subject to omitted variable biases...
- Let's make things concrete:
  - We have a sample of mothers with at least two children
  - $D_i = \mathbf{1}$(Having a third birth).
  - $Z_{i1} = \mathbf{1}$(Having twins in the second birth)
  - $Z_{i2} = \mathbf{1}$(First two births have the same gender)

Table 4.1.4: Wald estimates of labor supply effects

			IV Estimates using:			
			Twins		Sex-composition	
Dependent variable	Mean	OLS	First stage	Wald estimates	First stage	Wald estimates
	(1)	(2)	(3)	(4)	(5)	(6)
Employment	0.528	-0.167	0.625	-0.083	0.067	-0.135
		(0.002)	(0.011)	(0.017)	(0.002)	(0.029)
Weeks worked	19.0	-8.05	"	-3.83	"	-6.23
		(0.09)	"	(0.758)	"	(1.29)
Hours/week	16.7	-6.02	"	-3.39	"	-5.54
		(0.08)	"	(0.637)	"	(1.08)

Note: The table reports OLS and Wald estimates of the effects of a third birth on labor supply using twins and sex-composition instruments. Data are from the Angrist and Evans (1998) extract including married women aged 21-35 with at least two children in the 1980 Census. OLS models include controls for mother's age, age at first birth, dummies for the sex of first and second births, and dummies for race.

Table 4.4.3: Complier-characteristics ratios for twins and sex-composition instruments

Variable	$E[x]$ (1)	Twins at second birth $E[x\|D_1 > D_0]$ (2)	Twins at second birth $P[x\|D_1 > D_0]/P[X]$ (3)	First two children are same sex $E[x\|D_1 > D_0]$ (6)	First two children are same sex $P[x\|D_1 > D_0]/P[X]$ (5)
Age 30 or older at first birth	0.00291	0.00404	1.39 (0.0201)	0.00233	0.995 (0.374)
Black or hispanic	0.125	0.103	0.822 (0.00421)	0.102	0.814 (0.0775)
High school graduate	0.822	0.861	1.048 (0.000772)	0.815	0.998 (0.0140)
College graduate	0.132	0.151	1.14 (0.00376)	0.0904	0.704 (0.0692)

Notes: The table reports an analysis of complier characteristics for twins and sex-composition instruments. The ratios in columns 3 and 5 give the relative likelihood compliers have the characteristic indicated in each row. Data are from the 1980 Census 5% sample, including married mothers age 21-35 with at least two children, as in Angrist and Evans (1998). The sample size is 254,654 for all columns.

Testing exclusion

We can actually say something about exclusion even with only one instrument and no covariates

Suppose that the outcome, treatment, and instrument are all binary. We observe the treatment probabilities

$$Pr\left[T_i = 1 | Z_i = 1\right] = 0.9$$
$$Pr\left[T_i = 1 | Z_i = 0\right] = 0.8$$

and the outcome probabilities

$$Pr\left[Y_i = 1 | T_i = 1, Z_i = 1\right] = 0.2$$
$$Pr\left[Y_i = 1 | T_i = 1, Z_i = 0\right] = 0.8$$
$$Pr\left[Y_i = 1 | T_i = 0, Z_i = 1\right] = 0.35$$
$$Pr\left[Y_i = 1 | T_i = 0, Z_i = 0\right] = 0.4$$

Notice anything?

Testing exclusion

Based on the mixture table:

- What is the share of always takers, π_{at}?
- What is the share of compliers, π_c?
- What is the share of never takers, π_{nt}?

Testing exclusion

Based on the mixture table:

- What is the share of always takers, π_{at}?
- What is the share of compliers, π_c?
- What is the share of never takers, π_{nt}?

- Tell me how to identify $E[Y_i(1)|T_i(1) = 1, T_i(0) = 0]$
- Tell me what number you find for it
- Tell me what you think is going on

Testing exclusion

The first stage is too small to rationalize the difference between the always taker mean outcome and the complier/always taker mix

- There are not enough compliers to generate a drop from 0.8 to 0.2, even if all compliers have $Y_i(1) = 0$

This implies that T_i cannot be the only channel through which Z_i affects Y_i

Testing exclusion

More generally, let $p^{at} = \pi^{at}/(\pi^c + \pi^{at})$ be the fraction of compliers in the $T_i = Z_i = 1$ mixture. Then

$$E\left[Y_i(1)|T_i(1) = T_i(0) = 1\right] \leq E\left[Y_i|Y_i \geq Q_{1-p^{at}}(Y_i|T_i = Z_i = 1), T_i = Z_i = 1\right]$$
$$E\left[Y_i(1)|T_i(1) = T_i(0) = 1\right] \geq E\left[Y_i|Y_i \leq Q_{p^{at}}(Y_i|T_i = Z_i = 1), T_i = Z_i = 1\right]$$

- Under the most extreme possible assumptions, always takers compose either the upper p^{at} portion or the lower p^{at} portion of the mixture distribution
- This generates bounds on the always taker means that can be rationalized by the mixture

Testing exclusion

$$E\left[Y_i(1)|T_i(1) = T_i(0) = 1\right] \leq E\left[Y_i|Y_i \geq Q_{1-p^{at}}(Y_i|T_i = Z_i = 1), T_i = Z_i = 1\right]$$
$$E\left[Y_i(1)|T_i(1) = T_i(0) = 1\right] \geq E\left[Y_i|Y_i \leq Q_{p^{at}}(Y_i|T_i = Z_i = 1), T_i = Z_i = 1\right]$$

- But the always taker mean is also point identified as $E\left[Y_i|T_i = 1, Z_i = 0\right]$
- If the observed always taker mean is outside the bounds implied by the mixture, the exclusion restriction must be violated

Testing exclusion

A similar argument can be used to bound the never taker mean and compare the bounds to $E\left[Y_i | T_i = 0, Z_i = 1\right]$

- Huber and Mellace (2015) propose a formal test of the exclusion restriction based on this idea
- Kitagawa (2015) develops a related test that amounts to checking whether complier CDFs are non-decreasing
  - This paper provides a "sharp" result: the IV assumptions provide no other testable implications
- These tests are generalizations of the "no first stage sample" idea: there is a limit on how large the reduced form can be given the share of the population that complies
- Such tests may not be informative in practice, since the bounds will be wide unless the first stage is very small

Continuous treatments

Classic treatment is from Angrist and Imbens (1995)

- Treatment is ordered for $j = 0, 1, \ldots, J$
- Potential outcomes $Y(j)$
- Very natural for some things like prices, transfer amounts
- Less natural for treatments like (no post-secondary, trade school, university)

Single binary instrument Z:

- Monotonicity assumption: $Pr(D(1) \geq D(0)) = 1$ or $Pr(D(1) \leq D(0)) = 1$
- What does this mean?

Continuous treatments

Main result: the Wald estimand is a weighted average of average causal responses $Y(j) - Y(j-1)$:

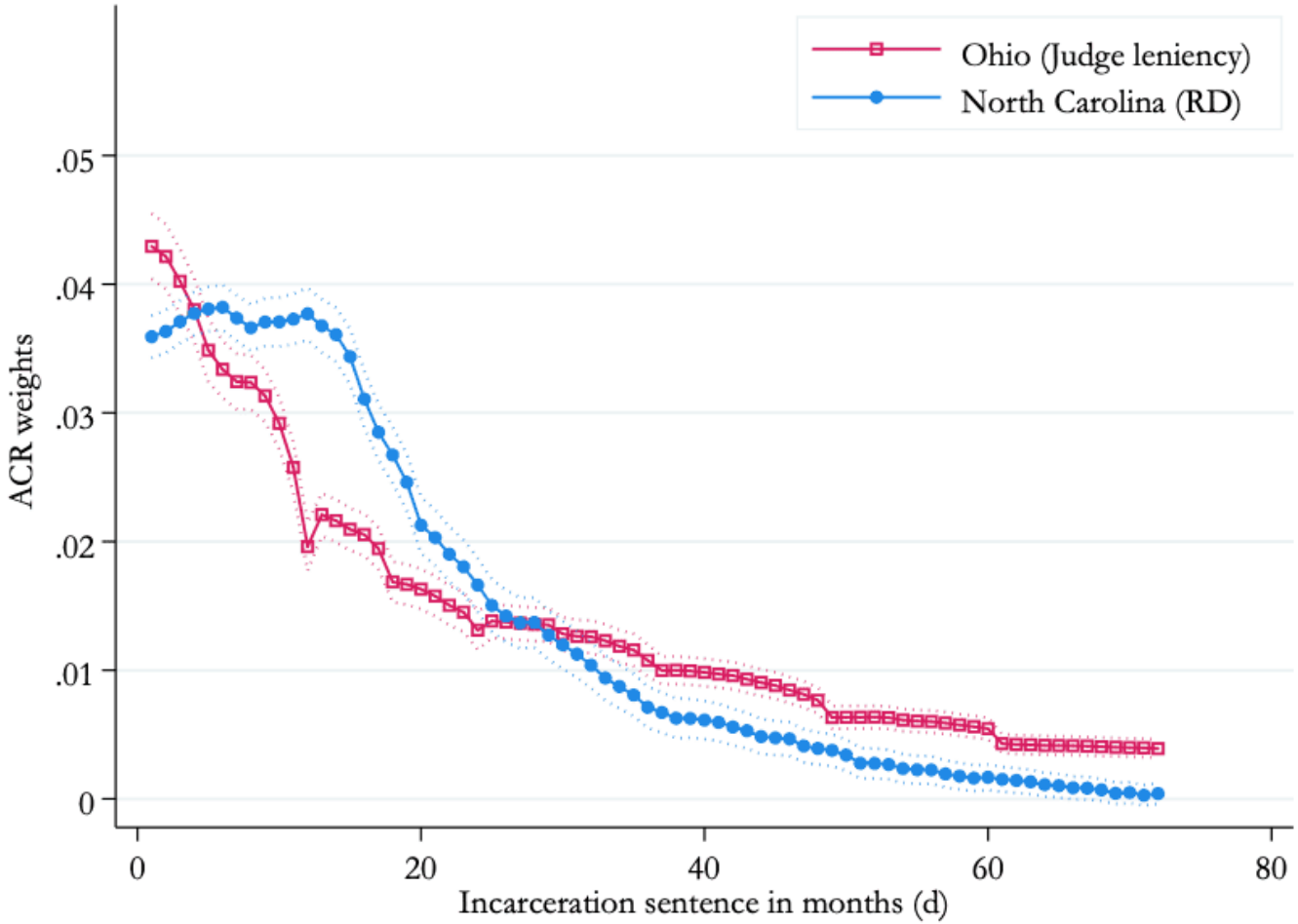
$$\frac{E[Y|Z{=}1] - E[Y|Z{=}0]}{E[D|Z{=}1] - E[D|Z{=}0]} = \sum_{j=1}^{J} \omega_j E[Y(j) - Y(j{-}1) \,|\, D(1) \geq j > D(0)]$$

$$\omega_j = \frac{P[D(1) \geq j > D(0)]}{\sum_{\ell=1}^{J} P[D(1) \geq \ell > D(0)]}$$

Note that ω can be estimated!

- Good practice to estimate/report the ACR weights
- Estimating complier means is actually harder (see Garin et al. 2024)

Figure B.4: Variation in incarceration induced by instruments



Continuous treatments and exclusion

Suppose that you have a continuous treatment like schooling S, and an instrument for it

- Assume that it satisfies the Angrist and Imbens (1995) conditions

Define a new treatment, $T_i \equiv 1[S \geq 12]$

- Can you instrument for that? Why or why not?

Continuous treatments and exclusion

Suppose that you have a continuous treatment like schooling S, and an instrument for it

- Assume that it satisfies the Angrist and Imbens (1995) conditions

Define a new treatment, $T_i \equiv 1[S \geq 12]$

- Can you instrument for that? Why or why not?

Requires assuming extensive margin compliers only (EMCO) (Rose and Shem-Tov, 2024)

- Some testable implications
- Remember: exclusion is instrument and treatment specific

Back to overid tests

Informed by our discussion of heterogeneous effects, let's return to overid tests

As discussed earlier, overid tests reject when different instruments yield different parameter estimates

- In a constant effects model, this implies that at least one must be invalid
- In a heterogeneous effect model, they may identify different LATEs

Can we distinguish between invalid instruments and treatment effect heterogeneity?

Back to overid tests

To fix ideas, return to a model with binary treatment T_i, binary instrument Z_i, and covariates X_i. Saturated 2SLS model:

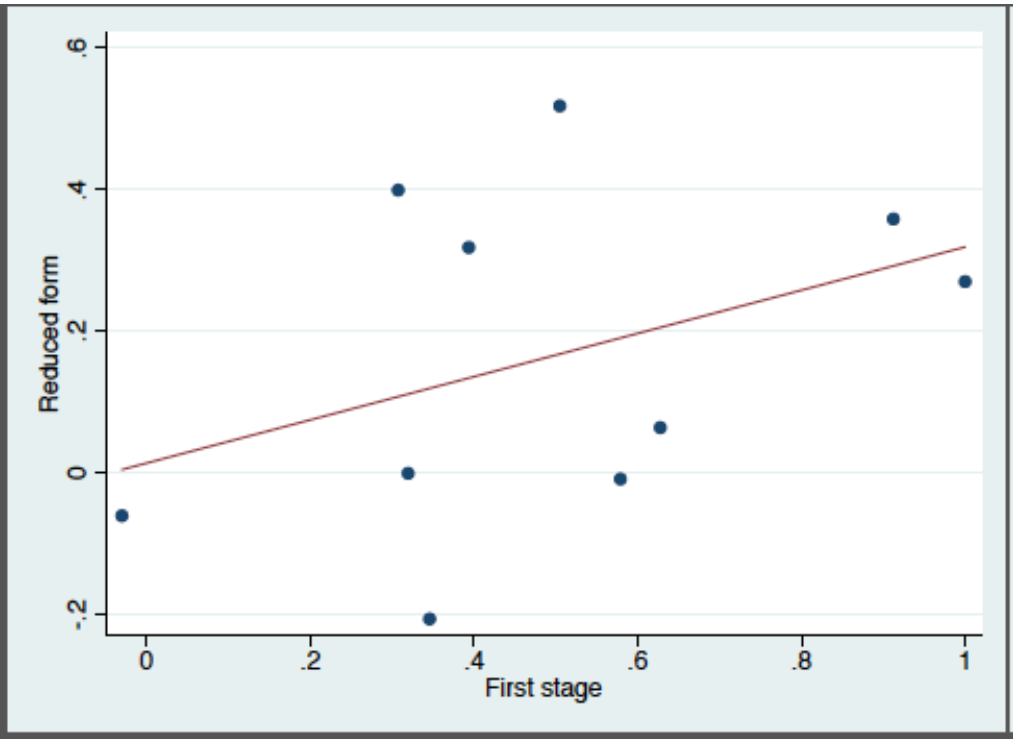
$$Y_i = X_i'\alpha + \rho T_i + \varepsilon_i$$
$$T_i = X_i'\lambda + Z_i \cdot X_i'\pi + \eta_i$$

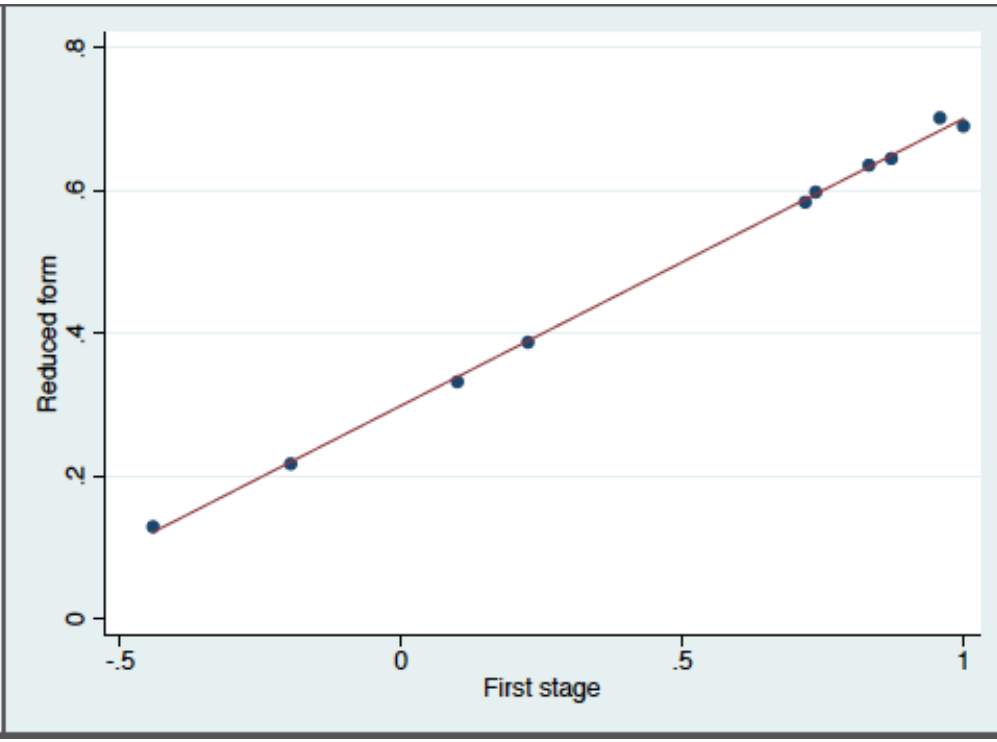
The overid test for this model checks whether Wald estimates are the same for each value of X_i

Covariate-specific reduced forms and first stages:

$$\delta(x) = E\left[Y_i | Z_i = 1, X_i = x\right] - E\left[Y_i | Z_i = 0, X_i = x\right]$$
$$\pi(x) = E\left[T_i | Z_i = 1, X_i = x\right] - E\left[T_i | Z_i = 0, X_i = x\right]$$

Suppose the overid test rejects, and consider two possible plots of $\hat{\delta}(x)$ against $\hat{\pi}(x)$


Reduced form
-.2
0
.2
.4
.6
0
.2
.4
.6
.8
1
First stage


Reduced form
0
.2
.4
.6
.8
-.5
0
.5
1
First stage

IV makes its way in clinical trials

Research question: do pre-cancer screenings improve health?

Doctors and patients can't be forced to test, so researchers have tried to answer this by randomizing the opportunity to screen

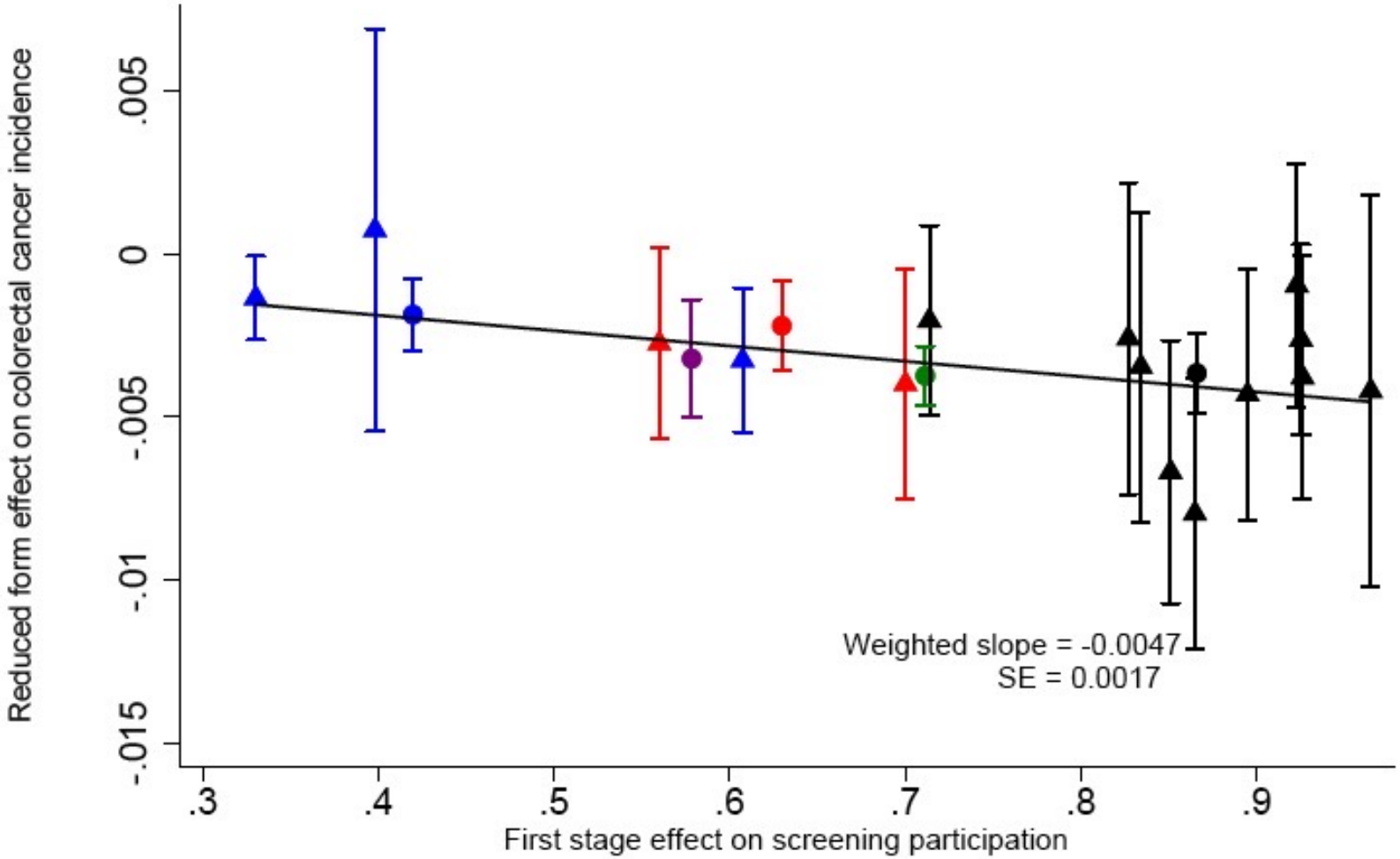
- Studies in this area thus report intention-to-treat (or intention-to-screen) effects: $\mathrm{E}[Y_i|Z_i = 1] - \mathrm{E}[Y_i|Z_i = 0]$
- There seems to be a lot of heterogeneity in the ITTs estimated across a whole variety of studies

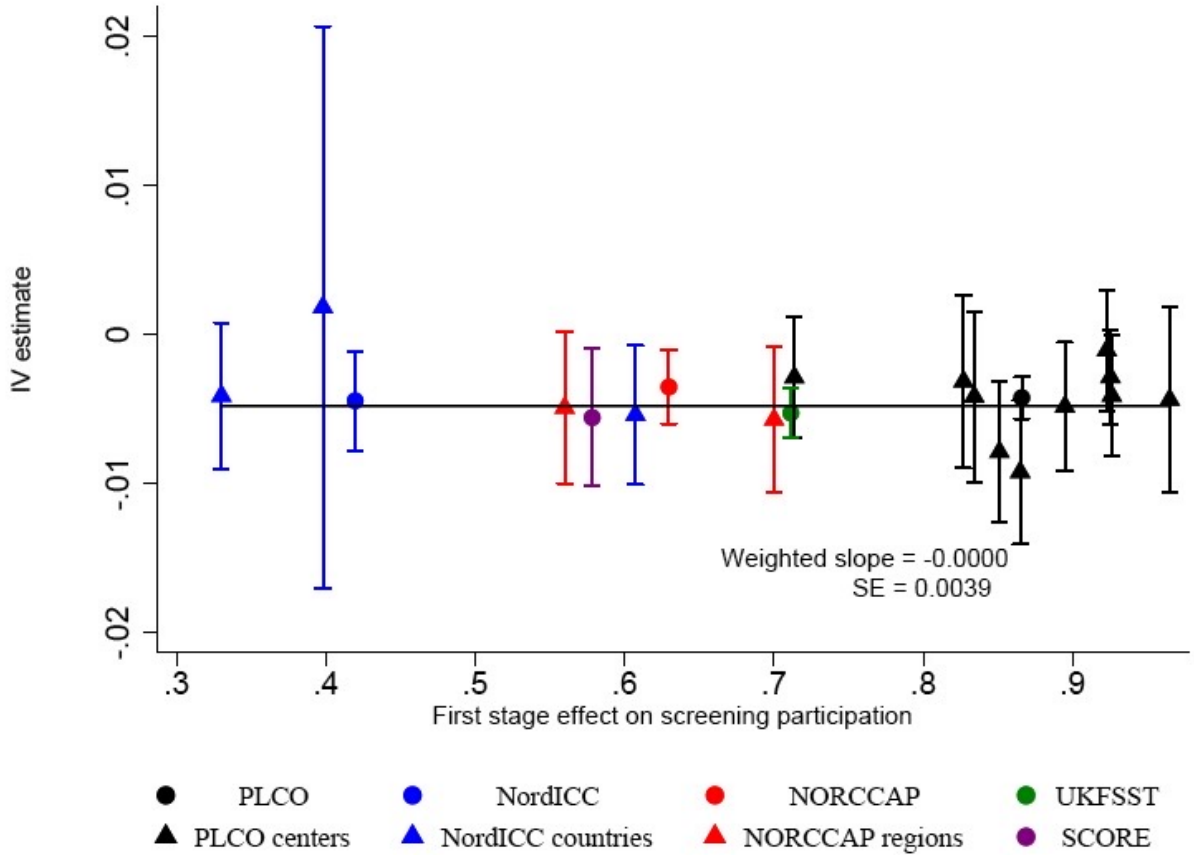
Angrist and Hull (2023) ask whether these intention-to-treat observations are applicable to other clinical environments

IV Methods Reconcile Intention-to-Screen Effects Across Pragmatic Cancer Screening Trials
Joshua Angrist and Peter Hull
NBER Working Paper No. 31443
July 2023, Revised September 2023
JEL No. C21,C26,C93,I12

ABSTRACT

Pragmatic cancer screening trials mimic real-world scenarios in which patients and doctors are the ultimate arbiters of treatment. Intention-to-screen (ITS) analyses of such trials maintain randomization-based apples-to-apples comparisons, but differential adherence (the failure of subjects assigned to screening to actually get screened) makes ITS effects hard to compare across trials and sites. We show how instrumental variables (IV) methods address the nonadherence challenge in a comparison of estimates from 17 sites in five randomized trials measuring screening effects on colorectal cancer incidence. While adherence rates and ITS estimates vary widely across and within trials, IV estimates of per-protocol screening effects are remarkably consistent. An application of simple IV tools, including graphical analysis and formal statistical tests, shows how differential adherence explains variation in ITS impact. Screening compliers are also shown to have demographic characteristics broadly similar to those of the full trial study sample. These findings argue for the clinical relevance of IV estimates of cancer screening effects.


Reduced form effect on colorectal cancer incidence
.005
0
-.005
-.01
-.015
.3
.4
.5
.6
.7
.8
.9
First stage effect on screening participation
Weighted slope = -0.0047
SE = 0.0017


IV estimate
.02
.01
0
-.01
-.02
.3
.4
.5
.6
.7
.8
.9
First stage effect on screening participation
Weighted slope = -0.0000
SE = 0.0039
PLCO
NordICC
NORCCAP
UKFSST
PLCO centers
NordICC countries
NORCCAP regions
SCORE

Final thoughts

IV is a powerful tool to deal with omitted variable bias

- Robust to heterogeneous effects

Often overlooked parts of thinking about an instrument:

- Who does my instrument affect (compliers / relationship to ITT/TOT/TNT/ATE)
- What does my instrument do (ACR weights / exclusion / future mechanisms)

APPENDIX

Original many/weak slides, kept for comparison with the rewritten section. Also contains the trace derivation of the Bekker approximation.

IV in Finite Samples

- OLS is an unbiased estimator of population projection coefficients in finite samples (with a linear CEF or nonstochastic regressors)
- IV estimators do not have this property
- IV is a ratio of reduced form and first stage coefficients:

$$E\left[\hat{\rho}_{IV}\right] = E\left[\frac{\hat{\delta}}{\hat{\pi}}\right]$$

$$\neq \frac{E\left[\hat{\delta}\right]}{E\left[\hat{\pi}\right]}$$

$$= \rho$$

IV in Finite Samples: Example

- Consider the following reduced form and first stage:

$$Y_i = \lambda + \delta Z_i + u_i$$

$$S_i = \kappa + \pi Z_i + \eta_i$$

- Suppose the instruments are non-stochastic, and

$$(u_i, \eta_i) \sim N\left(0, \begin{bmatrix} \sigma_u^2 & \tau\sigma_u\sigma_\eta \\ \tau\sigma_u\sigma_\eta & \sigma_\eta^2 \end{bmatrix}\right)$$

- Then the finite-sample distribution of the reduced form and first stage coefficients is

$$(\hat{\delta}, \hat{\pi}) \sim N\left((\delta, \pi), \begin{bmatrix} \sigma_u^2 \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} & \tau\sigma_u\sigma_\eta \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} \\ \tau\sigma_u\sigma_\eta \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} & \sigma_\eta^2 \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} \end{bmatrix}\right)$$

IV in Finite Samples: Example

- The IV coefficient, $\hat{\rho} = \hat{\delta}/\hat{\pi}$, is a ratio of two jointly normal random variables
- $\hat{\rho}$ therefore follows a normal ratio distribution
- Normal ratio distributions do not have moments, so $E[\hat{\rho}]$, $Var(\hat{\rho})$, and all higher moments do not exist. In this sense $\hat{\rho}$ is biased
- If $\tau = 0$ and $\pi \to 0$, $\hat{\rho}$ becomes the ratio of two independent mean zero normals, which follows a Cauchy distribution (see Nelson and Starz 1990 for more)

IV in Finite Samples: Many/Weak Problems

- In general, just-identified IV estimators have no moments and fat tails
- However, just-id estimators are approximately centered in the right place in the sense that $Q_{0.5}(\hat{\rho}_{IV}) \approx \rho$ (more details later)
- It turns out that overidentified IV estimators are not centered at the right place: they are biased towards OLS
- The severity of finite-sample bias increases with the degree of overfitting AND as the first-stage relationship between instruments and endogenous regressors gets weaker
- IV bias is therefore worse when the instruments are "many/weak"
- Bound, Jaeger and Baker (1995) showed that this problem can be important in practice, spawning a large literature on dealing with many/weak instruments

Bound, Jaeger and Baker (1995)

- Bound, Jaeger and Baker (1995) revisit the Angrist/Krueger QOB instrument
- The Wald estimates we studied are just-identified, so the many/weak critique does not apply (a common misconception)
- However, AK also estimate overidentified 2SLS models of the form

$$Y_i = \rho S_i + \sum_{b=1}^{50} \theta_b B_{ib} + \sum_{t=1}^{30} \psi_t T_{it} + \varepsilon_i$$

$$S_i = \sum_{q=1}^{3} \left[\sum_{b=1}^{50} \pi_{qb} B_{ib} Q_{iq} + \sum_{t=1}^{30} \tau_{qt} T_{it} Q_{iq} \right]$$

$$+ \sum_{b=1}^{50} \kappa_b B_{ib} + \sum_{t=1}^{30} \phi_t T_{it} + \eta_i$$

- Here B_{ib} is a dummy for birth in state b, T_{it} is a dummy for birth in year t, and Q_{iq} is a dummy for birth in quarter q

Bound, Jaeger and Baker (1995)

- The AK model uses 30 quarter/year and 150 quarter/state interactions as instruments, so is overidentified by 179
- The interactions are meant to capture differences in the effects of compulsory schooling laws across states and over time
- From the point of view of 2SLS asymptotics, adding the interactions can only be a plus: more instruments improve the first stage fit and drive down standard errors
- But the QOB first stage is small and many of the interactions are likely to be irrelevant, so this is a severe many/weak scenario
- Bound, Jaeger and Baker replace the QOB dummies with randomly-generated fake QOB variables that have no first stage by design

Bound, Jaeger and Baker (1995)

	2SLS (real QOB)	2SLS (fake QOB)	OLS
30 QOB*year	0.081	0.062	0.063
interactions	(0.016)	(0.037)	(0.000)
180 QOB*year and state	0.083	0.060	0.063
interactions	(0.009)	(0.015)	(0.000)

Bound, Jaeger and Baker (1995)

- The Bound, Jaeger and Baker results show that the AK 2SLS specification essentially reproduces OLS even if the instruments are junk unrelated to schooling
- Disturbingly, conventional IV standard errors give no indication of the extent of the problem
- The fake QOB 180-instrument model generates an estimate that looks precise despite the lack of a first stage
- How can we diagnose many/weak problems, and correct them if they are present?

Many/Weak Bias

- Consider a model with a single endogenous regressor and Q instruments, written in matrix form:

$$Y = \rho S + \varepsilon$$

$$S = Z\pi + \eta$$

- Remember that we can write $\hat{S} = P_Z S$
- We can thus re-write the 2SLS estimator of ρ (regression of Y on $\hat{S}$) as

$$\hat{\rho}_{2SLS} = \rho + (S' P_Z S)^{-1} S' P_Z \varepsilon$$

$$= \rho + (S' P_Z S)^{-1} (\pi' Z' + \eta') P_Z \varepsilon$$

$$= \rho + (S' P_Z S)^{-1} \pi' Z' \varepsilon + (S' P_Z S)^{-1} \eta' P_Z \varepsilon$$

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] = E\left[\left(S'P_Z S\right)^{-1}\pi' Z'\varepsilon\right] + E\left[\left(S'P_Z S\right)^{-1}\eta' P_Z\varepsilon\right]$$

- It's difficult to make further progress since we cannot pass expectations inside the inverse
- Bekker (1994): passing the expectations through provides an accurate approximation under group asymptotics that allow Q to grow with N
- Then the finite-sample bias of 2SLS can be approximated

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \left(E\left[S'P_Z S\right]\right)^{-1}\pi' E\left[Z'\varepsilon\right] + \left(E\left[S'P_Z S\right]\right)^{-1} E\left[\eta' P_Z\varepsilon\right]$$

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \left(E\left[S'P_Z S\right]\right)^{-1} \pi' E\left[Z'\varepsilon\right] + \left(E\left[S'P_Z S\right]\right)^{-1} E\left[\eta' P_Z \varepsilon\right]$$

- The first term is zero (why)?
- The finite-sample bias of 2SLS comes from the fact that the numerator of the second term is not zero:

$$E\left[\eta' P_Z \varepsilon\right] \neq 0$$

- What is the intuition for this?

Many/Weak Bias

- To make further progress, let's use some matrix tricks
  - A scalar equals its trace
  - Trace is invariant to cyclic permutations
  - Trace is a linear operator that passes through expectations

$$E\left[\eta' P_Z \varepsilon | Z\right] = E\left[tr\left(\eta' P_Z \varepsilon\right) | Z\right]$$

$$= E\left[tr\left(P_Z \varepsilon \eta'\right) | Z\right]$$

$$= tr\left(P_Z E\left[\varepsilon \eta' | Z\right]\right)$$

$$= \sigma_{\eta\varepsilon} tr\left(Z(Z'Z)^{-1}Z'\right)$$

$$= \sigma_{\eta\varepsilon} tr\left((Z'Z)^{-1}Z'Z\right)$$

$$= \sigma_{\eta\varepsilon} Q$$

Many/Weak Bias

- The denominator of the bias is

$$E\left[S'P_Z S\right] = E\left[(\pi' Z' + \eta')P_Z(Z\pi + \eta)\right]$$

$$= E\left[\pi' Z' Z\pi\right] + E\left[\eta' P_Z \eta\right]$$

- Applying the same trace tricks to the second term shows that $E\left[\eta' P_Z \eta\right] = \sigma_\eta^2 Q$
- Then

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\varepsilon} Q}{E\left[\pi' Z' Z\pi\right] + \sigma_\eta^2 Q}$$

$$= \frac{\sigma_{\eta\varepsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1 + E\left[\pi' Z' Z\pi\right] / (\sigma_\eta^2 Q)}$$

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\varepsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1 + E\left[\pi' Z' Z \pi\right] / (\sigma_\eta^2 Q)}$$

- Notice that $E\left[\pi' Z' Z \pi\right]$ is the explained sum of squares from the regression of S on Z, while σ_η^2 is the residual sum of squares divided by N
- Then

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\varepsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1 + F}$$

- Here F is the first stage (population) F-statistic for the joint significance of all the instruments

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] = \frac{\sigma_{\eta\varepsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1+F}$$

- Note also that when F goes to zero, the 2SLS bias becomes $\sigma_{\eta\varepsilon}/\sigma_{\eta}^2$
- With a zero first stage, $S = \eta$ so $\sigma_{\eta\varepsilon}/\sigma_{\eta}^2 = Cov(S_i, \varepsilon_i)/Var(S_i)$, the bias of OLS
- As the first stage shrinks to zero, all of the variance in the fitted values comes from the endogenous part of $\hat{S}$, and the bias of 2SLS equals the bias of OLS

Many/Weak In Practice

- In practice it is a good idea to look at the first stage F-statistic
- In models with exogenous controls this should be the partial F on the excluded instruments
- Stock and Staiger (1997) suggest that many weak typically becomes a problem when $F < 10$, though this is not a theorem
- Standard errors based on Bekker (1994) asymptotics are more accurate in many/weak scenarios (see also Hansen et al. 2008)

Many/Weak In Practice

- The Limited Information Maximum Likelihood (LIML) estimator coincides with 2SLS for just-id models and is approximately median unbiased for overidentified models
- A comparison of LIML and 2SLS can help determine whether many/weak is a problem (though conventional LIML standard errors are too small)
- LIML has some unattractive features, however: it has fat tails (no moments), and is inconsistent with heteroskedasticity
- There are other estimators (e.g. Fuller) that correct some of the deficiencies of LIML; see Stock and Yogo (2005)
- The Anderson-Rubin (AR) test provides a method for testing hypotheses about ρ that is robust to weak instruments, though it has low power when the instruments are weak

The Judges Design

- But when it sensible to talk about a research design with "many" instruments? $\rightarrow$ a research design that uses differences in the harshness/leniency of judges or other examiners is a research design that often has to deal with many/weak potential problems
- Examples: Kling (2006) on criminal sentencing, Doyle (2007) on foster care, Maestas et al. (2013) on disability insurance, Dobbie (2015) on bankruptcy
- If judges are randomly assigned, the identity of the judge is a plausible instrument (there are many fixed effects for judges!)

The Judges Design

- Common approach to the judges design: Define

$$T_i^* = \frac{1}{n_{j(i)} - 1} \sum_{k \neq i} 1\{j(k) = j(i)\} T_k$$

- T_i^* is the mean treatment rate (e.g. incarceration rate) for the judge assigned to i, $j(i)$, for all cases not involving i
- T_i^* is then used as an instrument for i's actual treatment T_i

The Judges Design: JIVE

- Note that T_i^* is the fitted value from a first stage regression of removal on all the judge dummies, leaving out i
- The instruments are implicitly a full set of judge dummies, which may lead to many/weak problems
- Using a leave-out mean changes the estimator from 2SLS to jackknife IV (JIVE), which is one solution to the many/weak problem but not necessarily the optimal one
- Conventional standard errors for JIVE are also incorrect in many/weak scenarios.
- Despite this, many papers use judges design given its implicit attractiveness plus the possibility to summarize nicely heterogenous treatment effects (see the marginal treatment effects literature pioneered by Heckman and Vytacil, 1999,2005).